

Extraction and Cleaning Protocols

After all screening is completed, we will begin data extraction. All data extraction is completed using the agreed upon template and each author completes their initial data extraction independently.

Ultimately, two independent authors extracted data from each of the included articles. Once data was extracted, the two authors responsible for extraction met and discussed each disagreement. The final data analyzed in our review is after settling all disagreements.

2.1 Extracting Data

Data extraction occurred using Excel and was coordinated in Covidence. These instructions detail what to do after you have logged into our project in Covidence and have your excel sheet open.

For an article that does not already have two people who have extracted data:

1. Tag the article with your name in Covidence.
2. Fill out each column in the extraction sheet with the value you retrieved from the article. In cases where the value is in a large table or figure, record the figure/table name as well.
 - a. For the author column: report the first author's name.
 - b. For the date columns: report the date the article was presented (if at a conference), not when it was posted online.
 - c. For the columns about code/models: include links.
 - d. Notes on where to *potentially* find some of the values:
 - i. Columns in the Data section should be found in the article in a "Data" or "Datasets" section. Sometimes this information is also found in a table or figure.
 - ii. Data/Code availability may sometimes be found in a "Data Availability" or "Supplemental Material" section.
 1. To ensure there is no associated code, also use a search engine and search for "[article title] code" and "[article title] git."
 - iii. Information about implementation and architecture may be found in the attached "Appendix" or "Supplemental Material."
 - iv. Certain values can sometimes be found by using searching the article (CTRL+F) with specific keywords. E.g., "Hz" can be used to find the frequency or "segment" may help find windowing/segmentation protocols.
 1. **NOTE:** This method is not extensive, you should only use it in a last resort and should rely on reading the article for this information first.
3. For any column where you cannot find the information in the article, record "N/A."
4. For the columns about replicability and reproducibility, record values according to your understanding of the article. Include your reasoning according to our code scheme.

2.2 Resolving Data Extraction Disagreements

Once two authors have completed data extraction for the same article, they will meet synchronously to resolve differences. Both authors should be able to see the final data extraction sheet and verify the information being recorded for analysis is agreed upon. Resolving disagreements oftentimes happened in batches of 10-15 articles at once.

For an article that two authors have independently extracted data from:

1. Start a new row in the final data table in Excel.
2. Verify both authors are discussing the correct article.

3. For each column in the extraction template, one author should state the column and value. The second author should verbally respond.
 - a. If the authors responsible for extraction agree: enter the agreed upon value in the final data extraction sheet.
 - b. If the authors responsible for extraction disagree:
 - i. Both authors state what their reported values were and where they found their information.
 - ii. Both authors open a copy of the original article and re-extract the information the other found.
 - iii. Both authors discuss which value they believe is correct and why.
 - iv. The value that is agreed upon is recorded in the final data extraction sheet.